

# Adnan Harun Dogan

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## EDUCATION

- 🎓 **Master of Science in Computer Engineering** Sep 2022 – Jan 2025  
Middle East Technical University (METU) Ankara, Turkey
- **Thesis:** [Incorporating Piecewise Linear Functions with Constant Regions in Backpropagation](#)
  - **Advisor:** Prof. Dr. [Sinan Kalkan](#), Assoc. Prof. Dr. [Emre Akbas](#)
- 🎓 **Bachelor of Science in Computer Engineering** Sep 2018 – Jul 2022  
Middle East Technical University (METU) Ankara, Turkey

## EXPERIENCE

- 🏢 **Graduate Intern** (Supervisor: Prof. Dr. [Christian Holz](#)) Mar 2025 – Oct 2025  
Sensing, Interaction, & Perception Laboratory (SIPLab), ETH Zürich Zürich, Switzerland
- Developed [learned frequency-weighted Kalman filters](#) that denoise the innovation spectrum under spectral supervision
  - Built [EEG multitaper-spectrum deep models](#) (ConvLSTM) for real-time VR cybersickness detection, +30% over baselines
- 🏢 **Research Assistant** (Supervisor: Prof. Dr. [Sinan Kalkan](#), Assoc. Prof. Dr. [Emre Akbas](#)) Feb 2023 – Present  
Image Processing and Pattern Recognition Laboratory (ImageLab), METU Ankara, Turkey
- Contributed to [Bucketed Ranking Loss](#), speeding up ranking-based training of DETR/transformer detectors by up to 6x
  - Developed optimal-transport (Sinkhorn-Knopp) and ranking-based loss formulations for end-to-end object detection
- 🏢 **Undergraduate Intern** (Supervisor: Assoc. Prof. Dr. [Erol Sahin](#), Assoc. Prof. Dr. [Bugra Koku](#)) Oct 2021 – Feb 2022  
Center for Artificial Intelligence and Robotics (ROMER), METU Ankara, Turkey
- Built a [self-driving stack](#) for a physical RC car, and trained RL agents for autonomous navigation task in ROS/Gym/PyTorch
  - Simulated a [Hector-quadrotor swarm](#) in ROS/Gazebo, approximating numerical solvers for multi-agent localization
- 🏢 **Undergraduate Intern** (Supervisor: Prof. Dr. [Mohammad Abdullah Al-Faruque](#)) Jun 2021 – Sep 2021  
Cyber-Physical Systems Laboratory (AICPS), University of California, Irvine Irvine, California, USA
- Built [multi-modal deep models](#) for beat-by-beat cardiac and sleep abnormality detection from physiological signals
  - Fused time-series and spectral features for arrhythmia classification, competitive in international challenges

## PUBLICATIONS

- Dogan, A. H.**, Celtik, S. O., Baran, O. U., & Alemdar, H. (2026). *Prediction-driven adaptive mesh refinement with graph neural networks for transonic airfoil flows*. [\[Under Review\]](#)
- Dogan, A. H.**, Deniz, M., Alemdar, H., & Baran, O. U. (2026). CoFINN: Conservation flux informed neural networks for physics problems governed by conservation laws. *Scientific Reports*. [arXiv](#)
- Dogan, A. H.**, Demirel, B. U., & Holz, C. (2026). Frequency-weighted neural Kalman filters. *IEEE/ICRA'26*. [arXiv](#) · [PDF](#) · [Code](#)
- Demirel, B. U., **Dogan, A. H.**, Rossie, J., Möbus, M., & Holz, C. (2025). Beyond subjectivity: Continuous cybersickness detection using EEG-based multitaper spectrum estimation. *IEEE/TVCG'25*. DOI · [arXiv](#) · [Code](#)
- Yavuz, F., Cam, B. C., **Dogan, A. H.**, Oksuz, K., Kalkan, S., & Akbas, E. (2024). Bucketed ranking-based losses for efficient training of object detectors. *ECCV'24*. DOI · [arXiv](#) · [PDF](#) · [Code](#)
- Amin, M. A., **Dogan, A. H.**, Kuru, E. S., Sever, Y., & Angin, P. (2024). Misuse detection and response for orchestrated microservices based software. *AINA'24*. DOI · [Data](#)
- Karagoz, P., Cekineli, R. F., **Dogan, A. H.**, Oktay, B., Ozturk, A. U., Tonay, T. S., & Tuncel, B. M. (2023). Enhancing underground built heritage analysis with text mining: A case study on Cappadocia. *CNR Edizioni 2023*. ISBN · [PDF](#)
- Sever, Y., & **Dogan, A. H.** (2023). A Kubernetes dataset for misuse detection. *ITU J-FET 2023*. DOI · [PDF](#) · [Code](#)
- Sever, Y., Ekinici, G., **Dogan, A. H.**, Alparslan, B., Gurbuz, A. S., Jabrayilov, V., & Angin, P. (2022). An empirical analysis of IDS approaches in container security. *IEEE/SRMC 2022*. [🏆 Best Paper Award](#) DOI
- Demirel, B. U., **Dogan, A. H.**, & Al-Faruque, M. A. (2021). Two might do: A beat-by-beat classification of cardiac abnormalities using deep learning and domain-specific features. *CinC 2021*. DOI · [PDF](#) · [Code](#)
- Buyukbas, E. B., **Dogan, A. H.**, Ozturk, A. U., & Karagoz, P. (2021). Explainability in irony detection. *DaWaK 2021*. DOI

## SCORES, HONORS, & AWARDS

University Entrance Exam: ranked 1466th of 2.27M candidates, OSYM, Turkey Aug 2017

## TECHNICAL SKILLS & RESEARCH INTERESTS

**Languages & Tools:** 🐍 Python/PyTorch, 📁 C/C++, 📦 ROS2, 🦀 Rust

**DevOps & HPC:** 🐙 GitHub, 🐳 Docker, Slurm (Workload Manager), Apptainer

**Differentiable Optimization:** Combinatorial Optimization, Optimal Transport, Ranking-Based Losses

**Deep Learning:** Computer Vision, Object Detection, Self-Supervised Learning, Deep State Space Models